

EDUCATION

the Hong Kong University of Science and Technology (Guangzhou)
Mphil (AI Thrust) | Advisor: Prof. Chengwei Qin and Prof. Renjing Xu

Sep. 2025 - Jun. 2027

Xiamen University

B.Econ. in Finance, IUEC (Average Score: 87.53/100 Rank: 7/40, Taught in English)
B.Sc. in Statistics, WISE (the Second Degree, Taught in English)

Sep. 2021 - Jun. 2025

PUBLICATIONS

LumiBench: Benchmarking Illumination Robustness in Robotic Manipulation Policies

Yiman Pang, Haoming Gu, Wenjie Zhang, **Yu Yan**, Xiangy Chen, Mingqian Zhou, Zhengtu Liang, Boyang Cai, Tao Wang, & Renjing Xu.

ICRA 2027 Conference Submission

Deep Search in Real-World Web Environments

Xiaomin Yu, Hongliang Chen, Zhe Huang, Shilin Yan, Shao-Rong Su, Xuqiao Ma, Jiwei Fang, Jiayu Chen, Feiyang Wang, **Yu Yan**, Yini Huang, Ruoxi Cheng, Chen Chen, Yuhui Zhang, Chen Liu, Jianbo Lin, Yijiang Li, Xiaobin Hu, Zhuoran Lu, Zhongqi Yue, Weishi Wang, & Chengwei Qin.

ICLR 2027 Conference Submission

Unified-MAS: Universally Generating Domain-Specific Nodes for Empowering Automatic Multi-Agent Systems

Hehai Lin, **Yu Yan**, Zixuan Wang, Bo Xu, Sudong Wang, WeiQuan Huang, Ruochen Zhao, Minzhi Li, & Chengwei Qin.

EMNLP 2026 Conference

RESEARCH EXPERIENCE

Humanoid Computing Lab (HCLab)

Guangdong, China

RESEARCH ASSISTANT | ADVISOR: PROF. RENJING XU

Dec. 2025 - Present

- LumiBench: Benchmarking Illumination Robustness in Robotic Manipulation Policies**

Introduction: We propose LumiBench, a physical benchmark for evaluating the *illumination robustness* of robotic manipulation policies across varying lighting conditions.

Contribution: Data collection for *3 real-robot tasks*; Process *5 policies* in the RoboTwin simulation system; Drew figures for the paper.

- The First Prize in ICRA 2026 WBCD (Vienna, Austria) Competition Track 4 Deformable Manipulation**

Introduction: The competition features real-world industrial *deformable manipulation* tasks, utilizing ARX robotic arms to tackle challenges like *T-shirt flattening*.

Method: Our framework fine-tunes π_0 -five using *1k / 10h* expert demonstrations at *45Hz*, integrating *Heuristic DAgger* for robust recovery. Deploying *5* inference steps with *exponential action-chunk smoothing* achieves a *90%+* success rate within *20s*.

Contribution: Data collection for the *dual-arm T-shirt flattening* task, model training and fine-tuning for π_0 -five.

Language and Intelligence Lab (LAI)

Guangdong, China

RESEARCH ASSISTANT | ADVISOR: PROF. CHENGWEI QIN

Sep. 2025 - Present

- Deep Search in Real-World Web Environments**

Introduction: We propose OpenDS-World, a real-world benchmark for evaluating the long-horizon, cross-source research capabilities of *Deep Search agents* across heterogeneous *Web environments*.

Contribution: Designed *50 multimodal multi-hop questions* for the benchmark; Handled *dataset cleaning* and *human check*.

- Unified-MAS: Universally Generating Domain-Specific Nodes for Empowering Automatic Multi-Agent Systems**

Introduction: We propose UnifiedMAS, an *automatic multi-agent framework* that decouples granular node implementation from topological orchestration via *offline node synthesis* for knowledge-intensive domains tasks.

Contribution: Ran DeepFund on real-world market data to collect agent trajectory datasets; Benchmarked *4 automatic multi-agent systems* (MAS-Zero, AFlow, ScoreFlow, MAS²) on *4 benchmarks* (TravelPlanner, HealthBench, J1Bench, DeepFund); drew figures for the paper.

Paradoox AI

Guangdong, China

RESEARCH ASSISTANT | ADVISOR: PROF. YUYU LUO

Mar 2026 - present

- InfoKnight: Proactively Forecasting Error Gaps for Online Blocking in Autonomous Agent Tool-Use**

Introduction: Evaluation, interpretability, and trustworthy AI: exploring the capability boundaries of LLM in the financial domain.

Contribution: By developing a proactive, small-model-guided *Contextual Bandit framework*, the model secures financial AI agents against *data-retrieval attacks* by evaluating the 'Error Gap' between predicted and actual tool outputs prior to execution.

Xiamen University

Fujian, China

RESEARCH ASSISTANT | ADVISORS: PROF. CHENDI WANG

Mar 2025 - Jun 2025

- TCAD: A Real-Time Framework for Stock Market Manipulation Detection Based on Deep Learning**

Introduction: Market Manipulation, Anomaly Detection, FinTech, Unsupervised Deep Learning.

Contribution: By integrating the *Transformer* architecture with *parallel CNNs* to construct the TCAD framework, the model effectively captures both long-term dependencies and multi-scale local features within stock time-series data.

Quantecon

RESEARCH ASSISTANT | ADVISORS: PROF. THOMAS SARGENT (NOBEL LAUREATE IN ECONOMICS)

Fujian, China

Oct 2024 - Aug 2025

- Collaborated on building **QuantEcon**, an open-source platform for economic modeling and computational code sharing.
- Supervised by **Matthew McKay** (Chief Developer, Australian National University).
- Developed *parsing mechanisms* to isolate natural language text, enabling lossless extraction and restoration of mixed content containing complex Jupyter Notebook structures, Markdown syntax, and LaTeX mathematical formulas.
- Assisted in *building a large model-driven translation pipeline* to replace traditional machine translation, enabling high-fidelity Chinese-English conversion of lecture notes.

WORK EXPERIENCE

DeepWisdom

META GPT TEAM | PRODUCT OPERATION-AI AGENTS FOCUS

Fujian, China

Mar 2025 - Jun 2025

- **Product Competitor Analysis:**
Introduction: Conducted in-depth analysis of over ten **AI coding tools (such as Cursor, GitHub Copilot, Manus, Perplexity)**.
Result: Raised community engagement by 30%, maintaining **MGX's top position** on the Product Hunt weekly leaderboard.
- **User Operation and Work Order Management:**
Introduction: Combined with the work order data of **MGX-BUG-Report**, a 24 - hour response SOP for user demand feedback was formed and incorporated into the company's overall product operation process.

GF Securities

EQUITY CAPITAL MARKETS (ECM)

Guangdong, China

Jan 2025 - Feb 2025

- **Roadshow Preparation:** Equity structure analysis and financial competitiveness evaluation. In-depth research on the wire and cable industry.
- **Report Drafting, Interview Coordination:** Report Drafting-Assisted in due diligence for IPO and refinancing projects.

EXTRACURRICULAR ACTIVITIES

- | | | |
|------|---|-----------------|
| 2026 | First Place , ICRA 2026 What Bimanuals Can Do (WBCD) Challenge | Vienna, Austria |
| 2024 | National First Prize , 18th "Challenge Cup" Competition | Fujian, China |
| 2024 | Provincial Award , Innovation and Entrepreneurial Training Program | Fujian, China |
| 2023 | Provincial Third Prize , 2023 National Collegiate Mathematical Contest in Modeling | Fujian, China |

SKILLS

- | | |
|----------------------------|------------------------|
| English Proficiency | CET-6: 601; IELTS: 6.5 |
| Technology Skills | Python, R, MATLAB |